

The accuracy is:

Scegli un'alternativa:

- ☒ a. The number of correct significant digits in approximating some quantity.
- ☐ b. The number of digits with which a number is expressed.
- ☐ c. None of the above.

Given two random variables X and Y , Bayes Theorem implies that $p(y|x) = \frac{p(x|y)p(y)}{p(x)}$ where:

Scegli un'alternativa:

- ☐ a. $p(x|y)$ is called prior distribution on x .
- ☐ b. $p(x|y)$ is called posterior distribution on y .
- ☒ c. $p(x|y)$ is called likelihood on y .

Given two random variables X and Y , Bayes Theorem implies that

$$p(y|x) = \frac{p(x|y)p(y)}{p(x)} \text{ where:}$$

Scegli un'alternativa:

- ☐ a. $p(y|x)$ is called prior distribution on x .
- ☐ b. $p(y|x)$ is called likelihood on y .
- ☒ c. $p(y|x)$ is called posterior distribution on y .

$$\begin{bmatrix} e^{x_1} & x_1 e^{x_2} \end{bmatrix} \begin{bmatrix} e^t \\ 1 \end{bmatrix} = \begin{bmatrix} e^t & e^t e^t \end{bmatrix} \begin{bmatrix} e^t \\ 1 \end{bmatrix} = e^t e^t + e^t e^t = e^{2t} + e^{2t} = 2e^{2t}$$

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1 e^{x_2}$, $g : \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (e^t, t)$, then, if $h(t) = f(g(t))$:

Scegli un'alternativa:

- ☐ a. $h'(t) = te^t$.
- ☒ b. $h'(t) = 2e^{2t}$.
- ☐ c. $h'(t) = e^{2t}(t + 1)$.

$$\begin{bmatrix} 2x_1 + x_2, & x_1 \end{bmatrix} \begin{bmatrix} \cos t \\ -\sin t \end{bmatrix} = \begin{bmatrix} 2\sin t + \cos t, & \sin t \end{bmatrix} \begin{bmatrix} \cos t \\ -\sin t \end{bmatrix} = 2\cos t \sin t + \cos^2 t - \sin^2 t$$

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1^2 + x_1x_2$, $g : \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (\sin(t), \cos(t))$,
then, if $h(t) = f(g(t))$:

Scegli un'alternativa:

- ☐ a. $h'(t) = \sin(2t) - \sin^2(t)$.
- ☐ b. $h'(t) = \sin(t) - \sin^2(2t)$.
- ☐ c. $h'(t) = \sin(t) \cos(t) - \sin^2(t)$.

$$\begin{bmatrix} 2x_1 & 2x_2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2x_2 & 2x_1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2x_1 & 2x_2 \end{bmatrix}$$

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1^2 + x_2^2$, $g : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $g(x_1, x_2) = (x_2, x_1)$,
then, if $h(x_1, x_2) = f(g(x_1, x_2))$:

Scegli un'alternativa:

- ☒ a. $\nabla h(x_1, x_2) = (2x_1, 2x_2)$.
- ☐ b. $\nabla h(x_1, x_2) = (2x_2, 2x_1)$.
- ☐ c. $\nabla h(x_1, x_2) = (1, 1)$.

$$\begin{bmatrix} e^{x_2} & x_1 e^{x_2} \end{bmatrix} \begin{bmatrix} 1 \\ 1/t \end{bmatrix} = \begin{bmatrix} e^{\log t} & t e^{\log t} \end{bmatrix} \begin{bmatrix} 1 \\ 1/t \end{bmatrix} = \begin{bmatrix} t & t^2 \end{bmatrix} \begin{bmatrix} 1 \\ 1/t \end{bmatrix} = t + t = 2t$$

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1 e^{x_2}$, $g : \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (t, \log t)$, then, if $h(t) = f(g(t))$:

Scegli un'alternativa:

- ☐ a. $h'(t) = t + 1$.
- ☐ b. $h'(t) = t^2 + 1$.
- ☒ c. $h'(t) = 2t$.

$$\begin{bmatrix} 2x_1 + x_2, x_1 \end{bmatrix} \begin{bmatrix} 2t \\ 1 \end{bmatrix} = \begin{bmatrix} 2t^2 + t, t^2 \end{bmatrix} \begin{bmatrix} 2t \\ 1 \end{bmatrix} = 4t^3 + 2t^2 + t^2 = 4t^3 + 3t^2$$

If $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1^2 + x_1x_2$, $g: \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (t^2, t)$, then, if $h(t) = f(g(t))$:

Scegli un'alternativa:

- ☐ a. $h'(t) = t(2t - 1)^2 + t.$ $t(4t^2 \cancel{- 4t}) \cancel{+ t} = 4t^3 - 4t^2$
- ☐ b. $h'(t) = 4t^2 + 2t + 1.$
- ☐ c. $h'(t) = t(2t + 1)^2 - 2t^2.$ $t(4t^2 + 4 + 4t) - 2t^2$

$$\begin{aligned}
 [2x_1, 2x_2] \begin{bmatrix} e^{x_2} & x_1 e^{x_1} \\ 0 & 1 \end{bmatrix} &= [2x_1 e^{x_2}, 2x_2] \begin{bmatrix} e^{x_2} & x_1 e^{x_2} \\ 0 & 1 \end{bmatrix} = \\
 &= [2x_1 e^{x_2} e^{x_2}, 2x_1 e^{x_2} x_1 e^{x_2} + 2x_2]
 \end{aligned}$$

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1^2 + x_2^2$, $g : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $g(x_1, x_2) = (x_1 e^{x_2}, x_2)$, then, if $h(x_1, x_2) = f(g(x_1, x_2))$:

Scegli un'alternativa:

- ☐ a. $\nabla h(x_1, x_2) = (2x_1 e^{x_2} (e^{x_2} + x_1), 2e^{x_2})$.
- ☐ b. $\nabla h(x_1, x_2) = (2x_1 e^{2x_2} (e^{x_1} + x_1), 2x_2)$.
- ☐ c. $\nabla h(x_1, x_2) = (2x_1 e^{x_2} (e^{x_2} + x_1), 2x_2)$.

$$\begin{bmatrix} x_2 & x_1 \end{bmatrix} \begin{bmatrix} 1 \\ 2t \end{bmatrix} = \begin{bmatrix} t^2 & t \end{bmatrix} \begin{bmatrix} 1 \\ 2t \end{bmatrix} = t^2 + 2t^2 = 3t^2$$

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1 x_2$, $g : \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (t, t^2)$, then, if $h(t) = f(g(t))$:

Scegli un'alternativa:

☒ a. $h'(t) = 3t^2$.

☐ b. $h'(t) = 3t^3$.

☐ c. $h'(t) = t^2$.

If

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

then:

Scegli un'alternativa:

☒ a. $K_2(A) = 1$.

☐ b. $K_2(A) = 4$.

☐ c. $K_2(A) = \frac{1}{2}$.

If

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

then:

$$\frac{4}{2}$$

Scegli un'alternativa:

- ☐ a. $K_2(A) = 4$.
- ☒ b. $K_2(A) = 2$.
- ☐ c. $K_2(A) = \frac{4}{3}$.

$$\frac{\|v - \tilde{v}\|_2}{\|v\|_2} = \frac{\sqrt{4^2 + 1^2}}{\sqrt{10^{17}}} = \sqrt{17} \cdot 10^{-6}$$

If vector $v = (10^6, 0)^T$ is approximated by vector $\tilde{v} = (999996, 1)^T$, then in $\|\cdot\|_2$ the relative error between v and $\tilde{v}$ is:

Scegli un'alternativa:

- ☒ a. $\sqrt{17} \cdot 10^{-6}$.
- ☐ b. None of the above.
- ☐ c. $4 \cdot 10^{-6}$.

If

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

then:

$$\frac{4}{1}$$

Scegli un'alternativa:

☐ a. $K_2(A) = \frac{1}{2}$.

☒ b. $K_2(A) = 4$.

☐ c. $K_2(A) = 2$.

A random variable $X : \Omega \rightarrow \mathcal{T}$ is continuous when:

Scegli un'alternativa:

- ☐ a. $\mathcal{T}$ is countable.
- ☒ b. $\mathcal{T} = \mathbb{R}$.
- ☐ c. Ω is continuous.

A random variable $X : \Omega \rightarrow \mathcal{T}$ is discrete when:

Scegli un'alternativa:

- ☐ a. $\mathcal{T} = \mathbb{R}$.
- ☐ b. Ω is countable.
- ☒ c. $\mathcal{T}$ is countable.

If

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 & 6 \end{bmatrix}$$

Then:

$$\cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 & 5 \end{bmatrix} \quad \lambda = 5$$

Scegli un'alternativa:

- ☐ a. $x = (1, 2)^T$ is an eigenvector of A .
- ☒ b. $x = (2, 1)^T$ is an eigenvector of A .
- ☐ c. $x = (0, 0)^T$ is an eigenvector of A .

If

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☒ a. $\lambda = 5$ is the eigenvalue associated with the eigenvector $x = (2, 1)^T$.
- ☐ b. $\lambda = 2$ is the eigenvalue associated with the eigenvector $x = (2, 1)^T$.
- ☐ c. $\lambda = 2$ is the eigenvalue associated with the eigenvector $x = (1, 2)^T$.

If

$$A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 0 \end{bmatrix}$$

Then:

$$\cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 0 \end{bmatrix} \quad \lambda = 2$$

Scegli un'alternativa:

- ☐ a. $x = (1, 1, 0)^T$ is an eigenvector of A .
- ☒ b. $x = (0, 1, 0)^T$ is an eigenvector of A .
- ☐ c. $x = (0, -1, 1)^T$ is an eigenvector of A .

If

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 & 0 \end{bmatrix} \quad \lambda = 4$$

Then:

Scegli un'alternativa:

- ☐ a. $x = (0, 0)^T$ is an eigenvector of A .
- ☒ b. $x = (1, 0)^T$ is an eigenvector of A .
- ☐ c. $x = (1, 1)^T$ is an eigenvector of A .

If

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

$\lambda = 2$

Then:

Scegli un'alternativa:

- ☒ a. $\lambda = 2$ is the eigenvalue associated with the eigenvector $x = (1, 0)^T$.
- ☐ b. $\lambda = 2$ is the eigenvalue associated with the eigenvector $x = (0, 1)^T$.
- ☐ c. $\lambda = 1$ is the eigenvalue associated with the eigenvector $x = (1, 0)^T$.

If $A \in \mathbb{R}^{n \times n}$, $x \in \mathbb{R}^n$ and

$$Ax = \lambda x$$

For $\lambda \in \mathbb{R}$, then:

Scegli un'alternativa:

- ☒ a. For any $c \in \mathbb{R}$, $c \neq 0$, cx is an eigenvector of A .
- ☐ b. cx is an eigenvector of A if and only if $c = 1$.
- ☐ c. None of the above.

If

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$\lambda = -1$

Then:

Scegli un'alternativa:

- ☐ a. $\lambda = 2$ is the eigenvalue associated with the eigenvector $x = (0, 1, 0)^T$.
- ☒ b. $\lambda = -1$ is the eigenvalue associated with the eigenvector $x = (0, 0, 1)^T$.
- ☐ c. $\lambda = 1$ is the eigenvalue associated with the eigenvector $x = (1, 0, 0)^T$.



In $\mathcal{F}(10, 2, -2, 2)$, if $x = \pi$, $y = e$, and $z = fl(x) - fl(y)$, then:

Scegli un'alternativa:

- ☐ a. $fl(z) = 0.43 \times 10^1$.
- ☐ b. $fl(z) = 0.44 \times 10^1$.
- ☐ c. $fl(z) = 0.40 \times 10^1$.

$$\begin{aligned}
 fl(x) &= 0,31 \cdot 10^1 \\
 fl(y) &= 0,27 \cdot 10^1 \\
 &= 0,31 \cdot 0,27 \cdot 10^2 = 0,0837 \cdot 10^2 = \\
 &= 0,837 \cdot 10^1 = \\
 &= 0,84 \cdot 10^1
 \end{aligned}$$

In $\mathcal{F}(10, 2, -2, 2)$, if $x = \pi$, $y = e$, and $z = fl(x) * fl(y)$, then:

Scegli un'alternativa:

- ☒ a. $fl(z) = 0.84 \times 10^1$.
- ☐ b. $fl(z) = 0.0837 \times 10^2$.
- ☐ c. $fl(z) = 0.837 \times 10^1$.

$$fl(x) = 0,192403 \cdot 10^3$$

$$fl(y) = 0,635782 \cdot 10^0$$

$$z = \begin{array}{r} 192,403 \\ 0,635782 \\ \hline 193,038782 \end{array}$$

$$fl(z) = 193,038 \xrightarrow{\text{with truncation}} 0,193039 \cdot 10^3$$

9 $\rightarrow$ with rounding

In $\mathcal{F}(10, 6, -3, 3)$, if $x = 192.403$, $y = 0.635782$, and $z = fl(x) + fl(y)$, then:

CHOOSE ROUNDING

Scegli un'alternativa:

- ☒ a. $fl(z) = 0.193039 \times 10^3$.
- ☐ b. $fl(z) = 0.193038 \times 10^3$.
- ☐ c. $fl(z) = 0.193038782 \times 10^3$.

$$fl(x) = 0,31 \cdot 10^1$$

$$fl(y) = 0,27 \cdot 10^1$$

In $\mathcal{F}(10, 2, -2, 2)$, if $x = \pi$, $y = e$, and $z = fl(x) + fl(y)$, then:

Scegli un'alternativa:

☐ a. $fl(z) = 0.585 \times 10^1$.

☒ b. $fl(z) = 0.58 \times 10^1$.

☐ c. $fl(z) = 0.59 \times 10^1$.

If $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $f \in \mathcal{C}^1(\mathbb{R}^n)$, then x^* is a minimum point if and only if:

Scegli un'alternativa:

- ☐ a. $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive semi-definite.
- ☐ b. $\nabla f(x^*) = 0$.
- ☒ c. $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite.

Gradient descent methods:

Scegli un'alternativa:

- ☐ a. If α is suitable chosen, $f \in C^1$, for any x_0 , always converges to a stationary point of $f(x)$.
- ☒ b. If α is suitable chosen, $f \in C^1$, for any x_0 , always converges to a minimum of $f(x)$.
- ☐ c. Always converges to a minimum of $f(x)$.

Gradient descent methods solves the optimization problem

$$\min_x f(x)$$

By:

Scegli un'alternativa:

- ☒ a. Generating a sequence $\{x_k\}_k$ such that, given x_0 , computes $x_{k+1} = x_k - \alpha \nabla f(x_k)$ for $\alpha > 0$ step-size.
- ☐ b. Generating a sequence $\{x_k\}_k$ such that, given x_0 , computes $x_{k+1} = x_k + \alpha \nabla f(x_k)$ for $\alpha > 0$ step-size.
- ☐ c. Generating a sequence $\{x_k\}_k$ such that, given x_0 , computes $x_{k+1} = x_k - \alpha \nabla f(x_k)$ for $\alpha \neq 0$ step-size.

$$x_1 = (1, 1) - \alpha(2, 2) = (1 - 2\alpha, 1 - 2\alpha) \quad (1 - 2\alpha)^2 < 1$$

$$|f(x_1)| < |f(x_2)| \quad \alpha(1 - 2\alpha)^2 < \alpha$$

$$|(1 - 2\alpha)^2 + (1 - 2\alpha)^2| < |1^2 + 1^2|$$

If $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1^2 + x_2^2$, then if the initial guess for a gradient descent iteration is $x^{(0)} = (1, 1)^T$ and $\alpha > 0$, then $|f(x^{(1)})| < |f(x^{(0)})|$ if:

Scegli un'alternativa:

☒ a. $0 < \alpha < 1$.

☐ b. $\alpha > 0$.

☐ c. $\alpha > \frac{1}{2}$.

$$K^2 < 1$$

$$-1 < K < 1$$

$$K > -1 \quad 1 - 2\alpha > -1 \quad \alpha < 1$$

$$K < 1 \quad 1 - 2\alpha < 1 \quad \alpha > 0$$

$$0 < \alpha < 1$$

$$x_1 = -\mathcal{L}(1, 0)$$

$$|f(x_1)| < |f(x_0)| \quad \mathcal{L} > 0$$

$$|e^{-\mathcal{L}}| < |1| \quad \log(e^{-\mathcal{L}}) < \log(1) \quad -\mathcal{L} < 0$$

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = e^{x_1} + x_2^2$, then if the initial guess for a gradient descent iteration is $x^{(0)} = (0, 0)^T$ and $\alpha > 0$, then $|f(x^{(1)})| < |f(x^{(0)})|$ if:

Scegli un'alternativa:

☐ a. $\alpha > \frac{1}{2}$.

☒ b. $\alpha > 0$.

☐ c. $0 < \alpha < 1$.

$$x_1 = (1, 1) - \frac{1}{2} (e, e)$$

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1 e^{x_2}$, then if the initial guess for a gradient descent iteration is $x^{(0)} = (1, 1)^T$ and $\alpha = \frac{1}{2}$, then:

Scegli un'alternativa:

- ☒ a. $x^{(1)} = (1 - \frac{e}{2}, 1 - \frac{e}{2})^T$.
- ☐ b. $x^{(1)} = (1 + \frac{e}{2}, 1 + \frac{e}{2})^T$.
- ☐ c. $x^{(1)} = (\frac{1}{2} - \frac{e}{2}, \frac{1}{2} - \frac{e}{2})^T$.

$$x_1 = - (1, 0)$$

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1 e^{x_2}$, then if the initial guess for a gradient descent iteration is $x^{(0)} = (0, 0)^T$ and $\alpha = 1$, then:

Scegli un'alternativa:

- ☐ a. $x^{(1)} = (1, 0)^T$.
- ☒ b. $x^{(1)} = (-1, 0)^T$.
- ☐ c. $x^{(1)} = (0, 0)^T$.

$$x_1 = (1, 1) - \mathcal{L}(2, 2)$$

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1^2 + x_2^2$, then if the initial guess for a gradient descent iteration is $x^{(0)} = (1, 1)^T$ and $\alpha > 0$, then:

Scegli un'alternativa:

- ☒ a. $x^{(1)} = (1 - 2\alpha, 1 - 2\alpha)^T$.
- ☐ b. $x^{(1)} = (1 - \alpha, 1 - \alpha)^T$.
- ☐ c. $x^{(1)} = (1 + 2\alpha, 1 + 2\alpha)^T$.

$$x_1 = -\alpha(1, 0)$$

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = e^{x_1} + x_2^2$, then if the initial guess for a gradient descent iteration is $x^{(0)} = (0, 0)^T$ and $\alpha > 0$, then:

Scegli un'alternativa:

- ☒ a. $x^{(1)} = (-\alpha, 0)^T$.
- ☐ b. $x^{(1)} = (0, 0)^T$.
- ☐ c. $x^{(1)} = (-\alpha, 2)^T$.

For Standard IEEE, double precision representation is:

Scegli un'alternativa:

- ☐ a. $\mathcal{F}(2, 64, -1024, 1023)$.
- ☐ b. None of the above.
- ☒ c. $\mathcal{F}(2, 53, -1024, 1023)$.

For Standard IEEE, single precision representation is:

Scegli un'alternativa:

- ☒ a. $\mathcal{F}(2, 24, -128, 127)$.
- ☐ b. None of the above.
- ☐ c. $\mathcal{F}(2, 32, -128, 127)$.

Given two independent random variables X and Y , then:

Scegli un'alternativa:

- ☐ a. $p(x) = p(y)$
- ☒ b. $p(y) = p(y|x)$
- ☐ c. $p(x|y) = p(y)$

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

Given two random variables X and Y , Bayes Theorem implies that:

Scegli un'alternativa:

- ☐ a. $p(x) = p(y)p(y|x) / p(y|x)$.
- ☐ b. $p(x) = p(x|y)p(y|x) / p(y)$.
- ☒ c. $p(y) = p(y|x)p(x) / p(x|y)$.

If $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $f(x) = \|Ax - b\|_2^2$ for $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, then:

Scegli un'alternativa:

- ☒ a. $\nabla f(x) = 2A^T(Ax - b)$.
- ☐ b. $\nabla f(x) = A^T(Ax - b)$.
- ☐ c. $\nabla f(x) = A(A^T x - b)$.

$$\nabla f(x) = 0$$

$$\cancel{A^T(Ax - b)} = 0 \quad A^T Ax - A^T b = 0$$

$$A^T Ax = A^T b$$



If $f : \mathbb{R}^n \rightarrow \mathbb{R}, f(x) = \|Ax - b\|_2^2$ for $A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m$, then the solution of $\nabla f(x) = 0$ is:

Scegli un'alternativa:

- ☐ a. $A^T Ax = b$
- ☐ b. $A^T Ax = A^T b$
- ☐ c. $Ax = b$.

The machine precision ϵ can be defined as:

Select one:

- ☐ a. The smallest number ϵ such that $fl(1 + \epsilon) = 1$.
- ☒ b. The smallest number ϵ such that $fl(1 + \epsilon) > 1$.
- ☐ c. None of the above.

Given two random variables X and Y such that $p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ and $p(y|x) = ce^{-|y-ax|}$, then the MAP reads:

Scegli un'alternativa:

- ☒ a. $x^* = \arg \min_x |y - ax| + \frac{1}{2}x^2$.
- ☐ b. $x^* = \arg \min_x |y - ax|$.
- ☐ c. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2$.

Given two random variables X and Y such that $p(x) = ce^{-|x|}$ and $p(y|x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(y-ax)^2}$, then the MAP reads:

Scegli un'alternativa:

- ☒ a. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + |x|.$
- ☐ b. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + \frac{1}{2}x^2.$
- ☐ c. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2.$

Given two random variables X and Y such that $p(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}x^2}$ and $p(y|x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(y-ax)^2}$, then the MAP reads:

Scegli un'alternativa:

- ☒ a. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + \frac{1}{2}x^2$.
- ☐ b. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + x^2$.
- ☐ c. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2$.

If

$$A = \begin{matrix} & A & \\ \begin{bmatrix} 2 & 1 & -2 \\ -1 & 0 & 1 \\ -1 & 2 & 1 \end{bmatrix} & \begin{matrix} A^T \\ \begin{bmatrix} 2 & -1 & -1 \\ 1 & 0 & 2 \\ -2 & 1 & 1 \end{bmatrix} \end{matrix} = \\ = \begin{bmatrix} 9 & \dots & \end{bmatrix} \neq I$$

Then:

Scegli un'alternativa:

- ☐ a. A is orthogonal.
- ☒ b. None of the above.
- ☐ c. A is symmetric and definite positive.

If $(2 - \lambda)(1 - \lambda) = 0$

$$\lambda = 2 \quad \lambda = 1$$

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☒ a. A is symmetric and positive definite.
- ☐ b. A is non-symmetric and not positive definite.
- ☐ c. A is symmetric but not positive definite.

If

$$A = \begin{matrix} & & A & & & & A^T \\ \begin{bmatrix} \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \end{bmatrix} & \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \end{bmatrix} & \begin{bmatrix} \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \end{bmatrix} & = \end{matrix}$$
$$= \begin{bmatrix} 9/9 & 0 & 0 \\ 0 & 9/9 & 0 \\ 0 & 0 & 9/9 \end{bmatrix} = I$$

Then:

Scegli un'alternativa:

- ☐ a. A is symmetric and definite positive.
- ☐ b. A is symmetric but not definite positive.
- ☒ c. A is orthogonal.

A definite positive if:

1) $\lambda_i > 0$

2) $\det(\text{Minors}(A)) > 0$

If

$$\det M_1 = 2 > 0$$

$$\det M_2 = -4 < 0 \rightarrow \text{not pos-def}$$

$$A = \begin{bmatrix} \boxed{2}^{M_1} & 2 & -1 \\ 2 & 0 & 2 \\ -1 & 2 & 3 \end{bmatrix}^{M_2}$$

Then:

Scegli un'alternativa:

- ☒ a. A is symmetric but not definite positive.
- ☐ b. A is symmetric and definite positive.
- ☐ c. A is orthogonal.

$$(-1-\lambda)(3-\lambda) = 0$$

If

$$A = \begin{bmatrix} -1 & 1 \\ 0 & 3 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. A is symmetric but not positive definite.
- ☐ b. A is symmetric and positive definite.
- ☒ c. A is non-symmetric and not positive definite.

$$(9-\lambda)(5-\lambda) - 36 = 0$$

If

$$(45 - 5\lambda - 9\lambda + \lambda^2) - 36 = 0 \quad \lambda^2 - 14\lambda + 9 = 0$$

$$A = \begin{bmatrix} 9 & 6 \\ 6 & 5 \end{bmatrix}$$

$$\lambda = \frac{7 \pm \sqrt{49 - 36}}{1} \begin{matrix} > 0 \\ > 0 \end{matrix}$$

Then:

Scegli un'alternativa:

- ☐ a. A is symmetric but not positive definite.
- ☐ b. A is non-symmetric and not positive definite.
- ☒ c. A is symmetric and positive definite.

If

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. A is symmetric and positive definite.
- ☒ b. A is symmetric but not positive definite.
- ☐ c. A is non-symmetric and not positive definite.

If

$$A = \begin{bmatrix} 2 & 1 & 2 \\ -2 & 2 & 1 \\ 1 & 2 & -2 \end{bmatrix} \begin{bmatrix} 2 & -2 & 1 \\ 1 & 2 & 2 \\ 2 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 9 & & \\ & \ddots & \\ & & \end{bmatrix} \neq I$$

Then:

Scegli un'alternativa:

- ☒ a. A is not orthogonal.
- ☐ b. A is orthogonal.
- ☐ c. A is symmetric but not definite positive.

$$\begin{array}{cccccccc}
 \begin{array}{l} 103 \\ 110 \\ -112 \end{array} & \rightarrow & \begin{array}{l} 103 \\ 022 \\ -112 \end{array} & \rightarrow & \begin{array}{l} 103 \\ 022 \\ 015 \end{array} & \rightarrow & \begin{array}{l} 103 \\ 011 \\ 015 \end{array} & \rightarrow & \begin{array}{l} 103 \\ 011 \\ 004 \end{array} & \rightarrow & \begin{array}{l} 103 \\ 011 \\ 001 \end{array} & \rightarrow & \begin{array}{l} 103 \\ 010 \\ 001 \end{array} & \rightarrow & \begin{array}{l} 100 \\ 010 \\ 001 \end{array}
 \end{array}$$

If

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 1 & 1 & 0 \\ -1 & 1 & 2 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. $\text{rank}(A) = 2$.
- ☐ b. $\text{rank}(A) = 1$.
- ☒ c. $\text{rank}(A) = 3$.

$$\begin{array}{ccccccc}
 068 & & 034 & & 034 & & 034 & & 0-40 & & 010 & & 010 \\
 140 & \rightarrow & 120 & \rightarrow & 120 & \rightarrow & 120 & \rightarrow & 120 & \rightarrow & 100 & \rightarrow & 100 \\
 103 & & 103 & & 0-28 & & 0-14 & & 0-14 & & 004 & & 004
 \end{array}$$

If

$$A = \begin{bmatrix} 0 & 6 & 8 \\ 2 & 4 & 0 \\ 1 & 0 & 8 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☒ a. $\text{rank}(A) = 3$.
- ☐ b. $\text{rank}(A) = 2$.
- ☐ c. $\text{rank}(A) = 1$.

If

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☒ a. $\text{rank}(A) = 2$.
- ☐ b. $\text{rank}(A) = 1$.
- ☐ c. $\text{rank}(A) = 3$.

If

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

then:

Scegli un'alternativa:

- ☒ a. $\text{rank}(A) = 3$.
- ☐ b. $\text{rank}(A) = 4$.
- ☐ c. $\text{rank}(A) = 2$.

If

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -2 & 2 \\ -1 & 1 & -1 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. $\text{rank}(A) = 3$.
- ☒ b. $\text{rank}(A) = 1$.
- ☐ c. $\text{rank}(A) = 2$.

If

$$A = \begin{bmatrix} 0 & 2 & -1 \\ 1 & 1 & 0 \\ 1 & 3 & -1 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☒ a. $\text{rank}(A) = 2$.
- ☐ b. $\text{rank}(A) = 3$.
- ☐ c. $\text{rank}(A) = 1$.

If

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

then:

Scegli un'alternativa:

- ☒ a. $\text{rank}(A) = 4$.
- ☐ b. $\text{rank}(A) = 2$.
- ☐ c. $\text{rank}(A) = 3$.

If

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 2 & 1 & 2 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. $\text{rank}(A) = 1$.
- ☐ b. $\text{rank}(A) = 3$.
- ☒ c. $\text{rank}(A) = 2$.

Given two random variables X and Y such that $p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ and $p(y|x) = ce^{-|y-ax|}$, then the MLE reads:

Scegli un'alternativa:

- ☐ a. $x^* = \arg \min_x |y - ax| + x^2$.
- ☐ b. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2$.
- ☒ c. $x^* = \arg \min_x |y - ax|$.

Given two random variables X and Y such that $p(x) = ce^{-|x|}$ and $p(y|x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y-ax)^2}$, then the MLE reads:

Scegli un'alternativa:

- ☐ a. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + |x|$.
- ☐ b. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + \frac{1}{2}x^2$.
- ☒ c. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2$.

Given two random variables X and Y such that $p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x}$ and $p(y|x) = ce^{-|y-ax|}$, then the MLE reads:

Scegli un'alternativa:

- ☐ a. $x^* = \arg \min_x |y - ax| + x^2$.
- ☒ b. $x^* = \arg \min_x |y - ax|$.
- ☐ c. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2$.

Given two random variables X and Y such that $p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ and $p(y|x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y-ax)^2}$, then the MLE reads:

Scegli un'alternativa:

☐ a. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + \frac{1}{2}x^2$.

☐ b. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + x^2$.

☒ c. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2$.

If

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

then:

Scegli un'alternativa:

- ☐ a. The 2-norm of A is $\|A\|_2 = 1$.
- ☐ b. The 2-norm of A is $\|A\|_2 = 0$.
- ☒ c. The 2-norm of A is $\|A\|_2 = 3$.

If

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

then:

Scegli un'alternativa:

- ☐ a. The 2-norm of A is $\|A\|_2 = 2$.
- ☒ b. The 2-norm of A is $\|A\|_2 = 4$.
- ☐ c. The 2-norm of A is $\|A\|_2 = 2$.

If

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

then:

Scegli un'alternativa:

☒ a. The 2-norm of A is $\|A\|_2 = 3$.

☐ b. The 2-norm of A is $\|A\|_2 = 0$.

☐ c. The 2-norm of A is $\|A\|_2 = 3$.

If A is an $n \times n$ matrix, then

Scegli un'alternativa:

☐ a. $\|A\|_1 = \sqrt{\sum_{i=1}^n \sum_{j=1}^n a_{i,j}^2}$.

☐ b. $\|A\|_1 = \rho(A^T A)$.

☒ c. None of the above.

If A is an $n \times n$ matrix, then

Scegli un'alternativa:

☐ a. None of the above.

☒ b. $\|A\|_2 = \rho(A^T A)$.

☐ c. $\|A\|_2 = \sqrt{\sum_{i=1}^n \sum_{j=1}^n a_{i,j}^2}$.

If A is an $n \times n$ matrix, then

Scegli un'alternativa:

☒ a. $\|A\|_F = \sqrt{\sum_{i=1}^n \sum_{j=1}^n a_{i,j}^2}.$

☐ b. None of the above.

☐ c. $\|A\|_F = \rho(A^T A).$

If $A \in \mathbb{R}^{m \times n}$, $\|A\|_p = 0$, then:

Scegli un'alternativa:

- ☒ a. $A = 0$.
- ☐ b. $\text{rank}(A) = 0$.
- ☐ c. A can be both equal or not equal to 0.

A matrix $A \in \mathbb{R}^{n \times n}$ is orthogonal if:

Scegli un'alternativa:

☐ a. $A^{-1}A = I = AA^{-1}.$

☒ b. $A^T A = I = AA^T.$

☐ c. $A = A^T.$

If $X : \Omega \rightarrow \mathcal{T}$ is a continuous random variable, then a function $p : \mathcal{T} \rightarrow \mathbb{R}_+$ can be the PDF of X if:

Scegli un'alternativa:

- ☒ a. $\int_{\mathcal{T}} p(x)dx = 1.$
- ☐ b. $\int_{\Omega} p(x)dx = 1.$
- ☐ c. $\int_{\mathcal{T}} p(x)dx < \infty.$

For a random variable $X : \Omega \rightarrow \mathcal{T}$ with $\mathbb{E}[X] = 0$, it holds:

Scegli un'alternativa:

- ☐ a. $\text{Var}(X) = \mathbb{E}[X]$.
- ☒ b. $\text{Var}(X) = \mathbb{E}[X^2]$.
- ☐ c. $\text{Var}(X) = 0$.

If $X : \Omega \rightarrow \mathcal{T}$ is a discrete random variable, then a function $f_X : \mathcal{T} \rightarrow [0, 1]$ can be the PDF of X if:

Scegli un'alternativa:

- ☐ a. $\sum_{i \in \Omega} f_X(i) = 1.$
- ☐ b. $\int_{\Omega} f_X(x) dx = 1.$
- ☒ c. $\sum_{i \in \mathcal{T}} f_X(i) = 1.$

Given a discrete random variable $X : \Omega \rightarrow \mathcal{T}$, with $\mathcal{T} = \{1, 2, \dots, 6\}$, and $f_X = \{\frac{1}{6}, \frac{1}{6}, \dots, \frac{1}{6}\}$, then:

Scegli un'alternativa:

☐ a. $\mathbb{E}[X] = 21$.

☒ b. $\mathbb{E}[X] = 3.5$.

☐ c. $\mathbb{E}[X] = \frac{1}{6}$.

$$\frac{1}{6} + \frac{2}{6} + \frac{3}{6} + \frac{4}{6} + \frac{5}{6} + \frac{6}{6} = \frac{21}{6}$$

Given a continuous random variable $X : \Omega \rightarrow \mathcal{T}$, with $\mathcal{T} = [0, 1]$, and $p(x) = 3x^2$ its PDF, then:

Scegli un'alternativa:

- ☐ a. $\mathbb{E}[X] = 2$.
- ☐ b. $\mathbb{E}[X] = 3$.
- ☒ c. $\mathbb{E}[X] = \frac{3}{4}$.

$$\int_0^1 3x^2 = 3 \left[\frac{x^3}{3} \right]_0^1$$

Given a continuous random variable $X : \Omega \rightarrow \mathcal{T}$, with $\mathcal{T} = [0, 1]$, and $p(x) = 2x$ its PDF, then:

$$\int x \cdot 2x = 2 \left[\frac{x^3}{3} \right]_0^1$$

Scegli un'alternativa:

- ☒ a. $\mathbb{E}[X] = \frac{2}{3}$.
- ☐ b. $\mathbb{E}[X] = 2$.
- ☐ c. $\mathbb{E}[X] = 1$.

If $X : \Omega \rightarrow \mathcal{T}$ is a continuous random variable with PDG $p : \mathcal{T} \rightarrow \mathbb{R}_+$, then:

Scegli un'alternativa:

- ☐ a. $\mathbb{E}[X] = \int_{\mathcal{T}} p(x)dx.$
- ☐ b. $\mathbb{E}[X] = \int_{\Omega} xp(x)dx.$
- ☒ c. $\mathbb{E}[X] = \int_{\mathcal{T}} xp(x)dx.$

If $X : \Omega \rightarrow \mathcal{T}$ is a continuous random variable, its Probability Density Function (PDF) $p_X(x)$ is defined to be:

Scegli un'alternativa:

- ☐ a. $P(X = x) = p_X(x)$.
- ☐ b. $P(X = x) = \int x p_X(x) dx$.
- ☒ c. $P(X \in A) = \int_A p_X(x) dx$.

If $X : \Omega \rightarrow \mathcal{T}$ is a continuous random variable, its Probability Density Function (PDF) $p_X(x)$ is:

Scegli un'alternativa:

- ☒ a. A function $p_X : \mathcal{T} \rightarrow \mathbb{R}_+$.
- ☐ b. A function $p_X : \mathcal{T} \rightarrow [0, 1]$.
- ☐ c. A function $p_X : \Omega \rightarrow [0, 1]$.

If $X : \Omega \rightarrow \mathcal{T}$ is a discrete random variable, its Probability Mass Function (PMF) f_x is:

Select one:

- ☒ a. A function $f_X : \mathcal{T} \rightarrow [0, 1]$.
- ☐ b. A function $f_X : \Omega \rightarrow [0, 1]$.
- ☐ c. A function $f_X : \mathcal{T} \rightarrow \mathbb{R}$.

Given two discrete random variable $X_1 : \Omega \rightarrow \mathcal{T}, X_2 : \Omega \rightarrow \mathcal{T}$ with $\mathcal{T} = \{1, 2, 3\}$, and $f_{X_1} = \{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}, f_{X_2} = \{\frac{1}{2}, \frac{1}{6}, \frac{1}{3}\}$ their PMF, then:

Scegli un'alternativa:

☒ a. $\mathbb{E}[X_1] < \mathbb{E}[X_2]$.

☐ b. $\mathbb{E}[X_1] = \mathbb{E}[X_2]$.

☐ c. $\mathbb{E}[X_1] > \mathbb{E}[X_2]$.

$$2 < 1 + \frac{1}{2} + \frac{1}{3} = \frac{6+3+2}{6} = \frac{11}{6}$$

If $X : \Omega \rightarrow \mathcal{T}$ is a discrete random variable, its Probability Mass Function (PMF) f_x is:

Scegli un'alternativa:

- ☐ a. $f_X(x) = \int P(x)dx.$
- ☐ b. $f_X(x) = P(X \in x).$
- ☒ c. $f_X(x) = P(X = x).$

If $X : \Omega \rightarrow \mathcal{T}$ is a discrete random variable with PMG $f_X : \mathcal{T} \rightarrow [0, 1]$, then:

Scegli un'alternativa:

☐ a. $\mathbb{E}[X] = \sum_{i \in \mathcal{T}} i.$

☐ b. $\mathbb{E}[X] = \sum_{i \in \mathcal{T}} i f_X(i).$

◀ Exam subscription November 2

Vai a...

Given a discrete random variable $X : \Omega \rightarrow \mathcal{T}$, with $\mathcal{T} = \{1, 2, 3\}$, and $f_X = \{\frac{1}{2}, \frac{1}{6}, \frac{1}{3}\}$ its PMF, then:

Scegli un'alternativa:

- ☐ a. $\mathbb{E}[X] = 6$.
- ☐ b. $\mathbb{E}[X] = 2$.
- ☐ c. $\mathbb{E}[X] = \frac{11}{6}$.

Given a discrete random variable $X : \Omega \rightarrow \mathcal{T}$, with $\mathcal{T} = \{1, 2, 3\}$, and $f_X = \{\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\}$ its PMF, then:

Scegli un'alternativa:

- ☐ a. $\mathbb{E}[X] = 6$.
- ☐ b. $\mathbb{E}[X] = 2$.
- ☐ c. $\mathbb{E}[X] = \frac{13}{6}$.

Given two discrete random variable $X_1 : \Omega \rightarrow \mathcal{T}$, $X_2 : \Omega \rightarrow \mathcal{T}$ with $\mathcal{T} = \{1, 2, 3\}$, and $f_{X_1} = \{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}$, $f_{X_2} = \{\frac{1}{2}, \frac{1}{6}, \frac{1}{3}\}$ their PMF, then:

Scegli un'alternativa:

- ☐ a. $\mathbb{E}[X_1] < \mathbb{E}[X_2]$.
- ☐ b. $\mathbb{E}[X_1] = \mathbb{E}[X_2]$.
- ☐ c. $\mathbb{E}[X_1] > \mathbb{E}[X_2]$.

Given two discrete random variable $X_1 : \Omega \rightarrow \mathcal{T}$, $X_2 : \Omega \rightarrow \mathcal{T}$ with $\mathcal{T} = \{1, 2, 3\}$, and $f_{X_1} = \{\frac{1}{2}, \frac{1}{6}, \frac{1}{3}\}$, $f_{X_2} = \{\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\}$ their PMF, then:

Scegli un'alternativa:

- ☐ a. $E[X_1] > E[X_2]$.
- ☐ b. $E[X_1] = E[X_2]$.
- ☐ c. $E[X_1] < E[X_2]$.

The precision is:

Scegli un'alternativa:

- ☐ a. None of the above.
- ☒ b. The number of digits with which a number is expressed.
- ☐ c. The number of correct significant digits in approximating some quantity.

A random variable X is:

Scegli un'alternativa:

- ☒ a. A function $X : \Omega \rightarrow \mathcal{T}$.
- ☐ b. A variable that returns random elements with known probability.
- ☐ c. A set that contains the possible outcomes of the experiment.

If Ω is the sample space, $\mathcal{A}$ is the event space and $\mathcal{T}$ is a subset of $\mathbb{R}$, a random variable X is:

Scegli un'alternativa:

- ☐ a. A function $X : \Omega \rightarrow \mathcal{A}$.
- ☐ b. A function $X : \mathcal{A} \rightarrow \mathcal{T}$.
- ☒ c. A function $X : \Omega \rightarrow \mathcal{T}$.

In normalized scientific notation and base $\beta = 10$, if $x = 2.71$, then:

Scegli un'alternativa:

- ☒ a. The mantissa of x is 0.271 and the exponential part is 10^1 .
- ☐ b. The mantissa of x is 2.71 and the exponential part is 10^0 .
- ☐ c. None of the above.

If $A = U\Sigma V^T$ is the SVD decomposition of $A \in \mathbb{R}^{m \times n}$, then a dyade $A_i = u_i v_i^T$ of A is:

Scegli un'alternativa:

$$m \times 1 \times 1 \times n$$

- ☐ a. None of the above.
- ☐ b. A vector of length mn that express some properties of A .
- ☒ c. A rank-1 matrix of dimension $m \times n$.

$$\begin{aligned}
 A^T A &= (U \Sigma V^T)^T (U \Sigma V^T) = \\
 &= V \Sigma^T U^T U \Sigma V^T = \\
 &= V \Sigma^2 V^T
 \end{aligned}$$

If $A = U \Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Scegli un'alternativa:

- ☐ a. $A^T A = V^T \Sigma^2 V$.
- ☐ b. $A^T A = U \Sigma^2 U^T$.
- ☒ c. $A^T A = V \Sigma^2 V^T$.

$$AA^T = U \Sigma V^T V \Sigma U^T$$

If $A = U\Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Scegli un'alternativa:

- ☐ a. The rows of V^T are eigenvectors of AA^T .
- ☒ b. The columns of U are eigenvectors of AA^T .
- ☐ c. None of the above

$$A^T A = (U \Sigma V^T)^T U \Sigma V^T$$

$$V \Sigma^T \cancel{U^T U} \Sigma V^T = V \Sigma^2 V^T$$

If $A = U \Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Scegli un'alternativa:

- ☒ a. The rows of V^T are eigenvectors of $A^T A$.
- ☐ b. None of the above
- ☐ c. The columns of U are eigenvectors of $A^T A$.

Given a matrix $A \in \mathbb{R}^{m \times n}$, $m > n$, with $r = \text{rank}(A)$, then:

Scegli un'alternativa:

- ☐ a. None of the above.
- ☒ b. It is always possible to write A as $U\Sigma V^T$, where $\Sigma \in \mathbb{R}^{m \times n}$ is diagonal, $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$ are orthogonals.
- ☐ c. It is possible to write A as $U\Sigma V^T$, where $\Sigma \in \mathbb{R}^{m \times n}$ is diagonal, $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$ are orthogonals if and only if $\text{rank}(A) = n$.

$$U \Sigma V^T V \Sigma^T U^T$$

If $A = U\Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Scegli un'alternativa:

☐ a. $AA^T = U^T \Sigma^2 U$.

☒ b. $AA^T = U \Sigma^2 U^T$.

☐ c. $AA^T = U \Sigma U^T$.

If $A = U\Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Scegli un'alternativa:

- ☒ a. The elements of the diagonal matrix Σ are the singular values of A , in decreasing order.
- ☐ b. The elements of the diagonal matrix Σ are the eigenvalues of A , in decreasing order.
- ☐ c. None of the above.

If $A = U\Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Scegli un'alternativa:

- ☒ a. The singular values σ_i of A are $\sigma_i = \sqrt{\lambda_i(A^T A)}$ where $\lambda_i(A^T A)$ are the eigenvalues of $A^T A$.
- ☐ b. None of the above
- ☐ c. The singular values σ_i of A are $\sigma_i = \lambda_i(A^T A)$ where $\lambda_i(A^T A)$ are the eigenvalues of $A^T A$.

If $A = U\Sigma V^T$ is the SVD decomposition of $A \in \mathbb{R}^{m \times n}$, then the rank k approximation of A is:

Scegli un'alternativa:

- ☐ a. $\hat{A}(k) = \sum_{i=1}^n \sigma_i A_i$, where $A_i = u_i v_i^T$ is a dyade.
- ☐ b. $\hat{A}(k) = \sum_{i=1}^k A_i$, where $A_i = u_i v_i^T$ is a dyade.
- ☒ c. $\hat{A}(k) = \sum_{i=1}^k \sigma_i A_i$, where $A_i = u_i v_i^T$ is a dyade.

If $A = U\Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Scegli un'alternativa:

- ☐ a. The elements on the diagonal of Σ are strictly positive.
- ☐ b. None of the above.
- ☒ c. The elements of the diagonal matrix Σ are non-negative.

If $A = U\Sigma V^T$ is the SVD decomposition of $A \in \mathbb{R}^{m \times n}$, then its rank k approximation of $\hat{A}(k)$ satisfies:

Select one:

- ☒ a. $\hat{A}(k) = \arg \min_{rk(B)=k} \|A - B\|_2.$
- ☐ b. $\hat{A}(k) = \arg \min_{rk(B)=k} \|A - B\|_F.$
- ☐ c. $\hat{A}(k) = \sigma_{k+1}.$

If $A \in \mathbb{R}^{m \times n}$, then:

Scegli un'alternativa:

- ☒ a. $A^T A$ is symmetric but not necessarily positive definite.
- ☐ b. $A^T A$ is always symmetric and positive definite.
- ☐ c. It depends on A .

For a random variable $X : \Omega \rightarrow \mathcal{T}$, its variance is defined as:

Scegli un'alternativa:

- ☐ a. $Var(X) = \mathbb{E}[X - \mathbb{E}[X]]$.
- ☒ b. $Var(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$.
- ☐ c. $Var(X) = \mathbb{E}[X^2 - \mathbb{E}[X]^2]$.

For a random variable $X : \Omega \rightarrow \mathcal{T}$, it holds:

Scegli un'alternativa:

- ☒ a. $Var(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.
- ☐ b. $Var(X) = \mathbb{E}[X^2] + \mathbb{E}[X]^2$.
- ☐ c. $Var(X) = \mathbb{E}[X]^2 - \mathbb{E}[X^2]$.

For a random variable $X : \Omega \rightarrow \mathcal{T}$ with $\mathbb{E}[X] = 0$, it holds:

Scegli un'alternativa:

- ☒ a. $\text{Var}(X) = \mathbb{E}[X^2]$.
- ☐ b. $\text{Var}(X) = 0$.
- ☐ c. $\text{Var}(X) = \mathbb{E}[X]$.

$$\frac{\|v - \tilde{v}\|_1}{\|v\|_1} = \frac{\|(4, 1)\|_1}{\|(10^6, 0)\|_1} = \frac{4+1}{10^6} = 5 \cdot 10^{-6}$$

If vector $v = (10^6, 0)^T$ is approximated by vector $\tilde{v} = (999996, 1)^T$, then in $\|\cdot\|_1$ the relative error between v and $\tilde{v}$ is:

Scegli un'alternativa:

- ☐ a. $4 \cdot 10^{-6}$.
- ☐ b. None of the above.
- ☒ c. $5 \cdot 10^{-6}$.

If vector $v = (10^6, 0)^T$ is approximated by vector $\tilde{v} = (999996, 1)^T$, then in $\|\cdot\|_2$ the relative error between v and $\tilde{v}$ is:

Scegli un'alternativa:

- ☒ a. $\sqrt{17} \cdot 10^{-6}$.
- ☐ b. $4 \cdot 10^{-6}$.
- ☐ c. None of the above.

$$\frac{\|v - \tilde{v}\|_1}{\|v\|_1} = \frac{\|(0.1, -0.1)\|_1}{\|(1, 0)\|_1} = \frac{0.2}{1}$$

If vector $v = (1, 0)^T$ is approximated by vector $\tilde{v} = (1.1, 0.1)^T$, then in $\|\cdot\|_1$ the relative error between v and $\tilde{v}$ is:

Scegli un'alternativa:

- ☐ a. 0.1.
- ☒ b. 0.2.
- ☐ c. None of the above.

$$\frac{\|v - \tilde{v}\|_1}{\|v\|_1} = \frac{\|(4, 0)\|_1}{\|(10^6, 1)\|_1} = \frac{4}{10^6}$$

If vector $v = (10^6, 1)^T$ is approximated by vector $\tilde{v} = (999996, 1)^T$, then in $\|\cdot\|_\infty$ the relative error between v and $\tilde{v}$ is:

Scegli un'alternativa:

- ☐ a. 4.
- ☒ b. $4 \cdot 10^{-6}$.
- ☐ c. None of the above.

Given $x = (1, -1)^T$ and $y = (2, 1)^T$, then:

Scegli un'alternativa:

☒ a. None of the above.

☐ b. $x \perp y$.

☐ c. $x \parallel y$.

$$1 \cdot 2 - 1 \neq 0 \rightarrow \text{X}$$

$$1, -1 \neq \lambda (2, 1) \quad \text{X} \rightarrow \text{X}$$

$$1 \cdot 1 - 1 = 0$$

Given $x = (1, -1)^T$ and $y = (1, 1)^T$, then:

Scegli un'alternativa:

☒ a. $x \perp y$.

☐ b. None of the above.

☐ c. $x \parallel y$.